

FPGA Development for Radar, Radio-Astronomy and Communications

THE
RADAR
MASTERS COURSE



Dept. Electrical Engineering, University of Cape Town
Private Bag, Rondebosch, 7701, South Africa
<http://www.rmsg.uct.ac.za>



Presented by Dr John-Philip Taylor
Convened by Dr Stephen Paine

Day 4 – 6 August 2026

Outline

Injection Testing

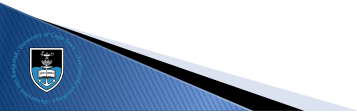
Advanced Simulation

Mutual Exclusion and Arbitration

Caching Systems

Numerically-controlled Oscillator

Pulse-width Modulation



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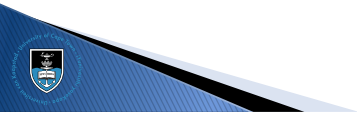
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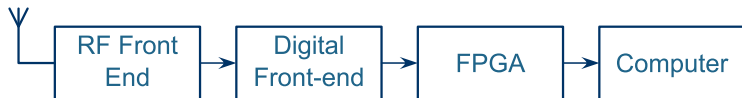
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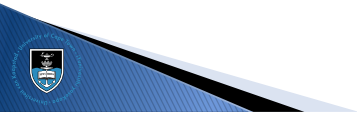
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Typical Development Challenges



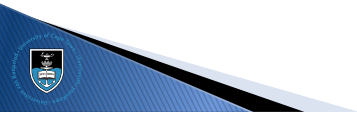
- ▶ The hardware (RF front-end and DFE) is not available at the time of FPGA firmware development
- ▶ There is no real-world data available
- ▶ Lab-generated signals are not suitable for DSP-chain testing
- ▶ Field tests are short-term experiments and expensive to run
- ▶ The DSP algorithms are experimental and uncertain



Early Development



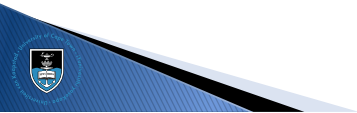
- ▶ The system is simulated
- ▶ The simulated ADC data is injected into the FPGA
- ▶ The DSP-chain is developed on this simulated data
- ▶ At first, the computer-based DSP does most, if not all, of the processing (FPGA firmware consist only of supporting infrastructure)
- ▶ As the algorithms mature, the functionality is moved into the FPGA



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Late Development

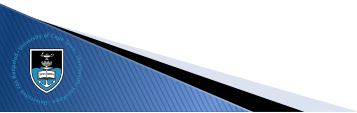


- ▶ The front-end hardware is built, but must be tested
- ▶ An FPGA-based logger is implemented that saves raw ADC data from a field trial
- ▶ This recorded data is analysed in order to test the hardware and injected into the FPGA to test the DSP-chain
- ▶ This same recorded data can be used in simulation...

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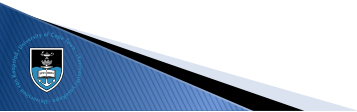
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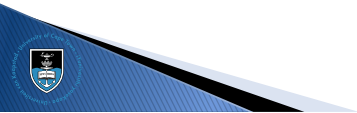
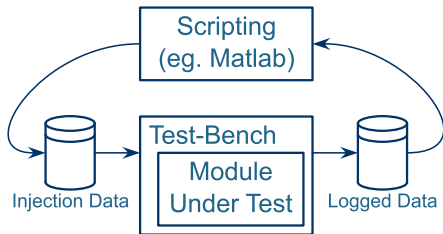
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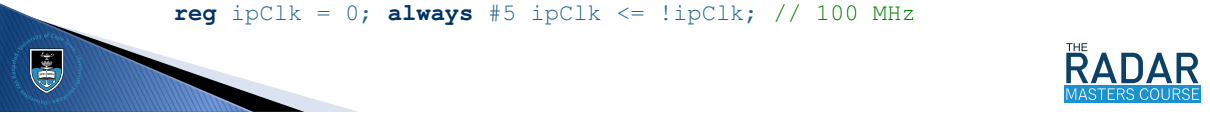
File I/O

```
`timescale 1ns/1ns
module File_IO_TB;

integer Input_File;
integer Output_File;

initial begin
    Input_File = $fopen("Test_File.txt", "rb");
    Output_File = $fopen("Output.txt" , "wb");
    $fwrite(
        Output_File,
        "Time [ns], Data [Hex], Data [Binary]\n"
    );
end

reg ipClk = 0; always #5 ipClk <= !ipClk; // 100 MHz
```



File I/O

```
reg [7:0]Byte; reg [15:0]Data; integer Result;

always @(posedge ipClk) begin
    Result = $fread(Byte, Input_File); Data[ 7:0] = Byte;
    Result = $fread(Byte, Input_File); Data[15:8] = Byte;

    if(Result < 1) begin
        $fclose(Input_File); $fclose(Output_File); $stop;
    end

    $fwrite(Output_File, "%d, %04X, %s\n", $time, Data, Data);
end

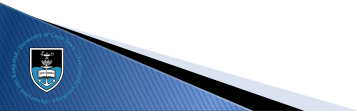
endmodule
```



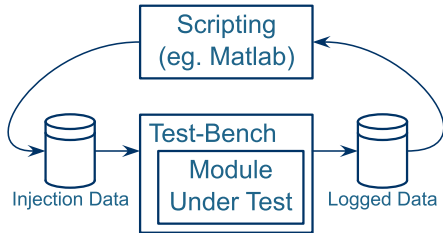
File I/O

Consult the Verilog-2001 standard:

- ▶ Section 17.2.4: “Reading data from a file”
- ▶ Section 17.2.8: “Loading memory data from a file”
- ▶ `$fread` reads any size register, but in big-endian
- ▶ `$fscanf` reads formatted data, including raw binary
- ▶ `$fwrite` works similar to `fprintf` from C, except that `%u` writes raw binary data
- ▶ `$fmonitor` logs to file whenever the signals change

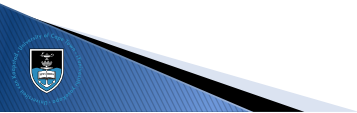


Data Injection (ADXL345_TB)



- Create a test file "ADXL345_TB.dat":

0123456789ABCDEFGHIJKLMN OPQRSTUVWXYZ



Data Injection (ADXL345_TB)

```
integer    File, n;
reg [15:0] Data;

initial begin
    File = $fopen("../Peripherals/ADXL345_TB.dat", "rb");

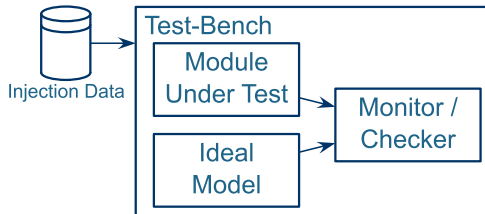
    for(n = 0; n < 16; n = n + 1) @(negedge SClk);
    forever begin
        for(n = 0; n < 8; n = n + 1) @(negedge SClk);
        $fread(Data, File);
        for(n = 0; n < 16; n = n + 1) begin
            @(negedge SClk);
            #40 {SDO, Data[15:1]} <= Data;
        end
    end
end
```



Data Injection (ADXL345_TB)



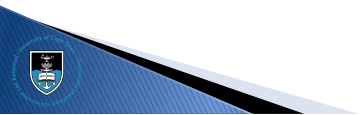
Automated Test-benches



- ▶ Very useful, but time-consuming to write...
- ▶ The “Ideal Model” can be read from a file, or calculated in the test-bench
- ▶ The “Monitor” issues errors (with `$error`), warnings (with `$warning`) or information (with `$info`) as appropriate
- ▶ Refer to the [Universal Verification Methodology](#) for more information



Coffee Break...



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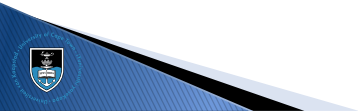
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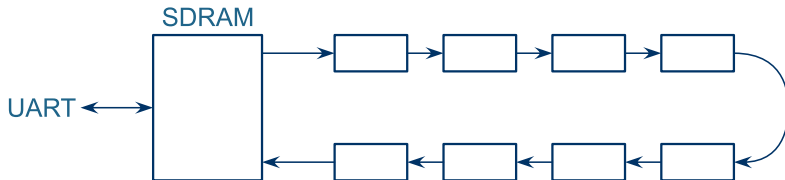
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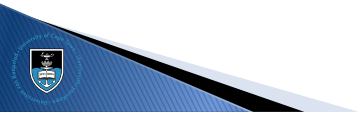
Pulse-width Modulation



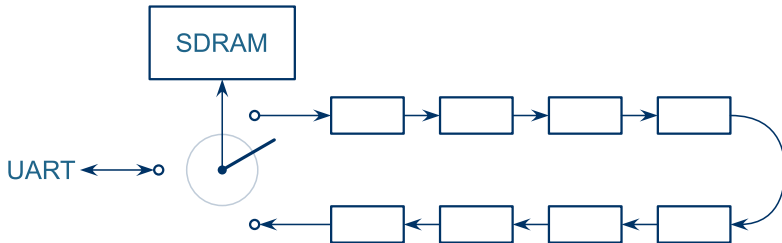
Sharing Resources



- ▶ Often many modules need access to the same resource, but only one module can interface with it at a time
- ▶ Examples include:
 - ▶ External memory (SDRAM, SD-card, etc.)
 - ▶ I²C bus
 - ▶ etc.

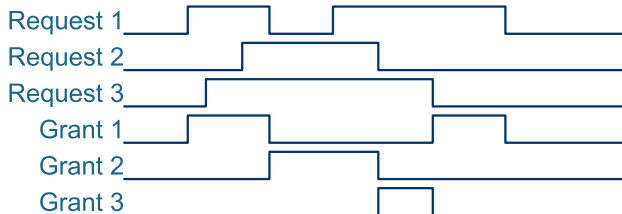


Mutual Exclusion



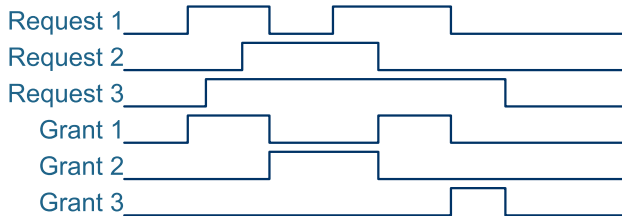
- ▶ Uses “Request” and “Grant” control lines
- ▶ The module would raise a request and then wait until it has been granted access before using the resource
- ▶ All the request lines go to a central control module that administers the granting of the resource

Round-robin Mutual Exclusion



- ▶ The requests are serviced in a circular order
- ▶ Guaranteed no starvation
- ▶ Does not scale well (in the case of many modules, many clock-cycles must be wasted checking each module's request line)

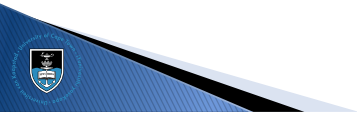
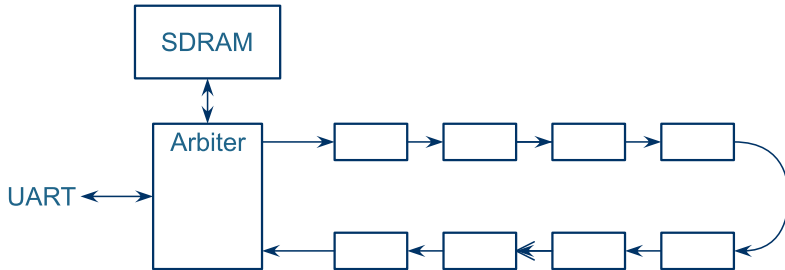
Priority-based Mutual Exclusion



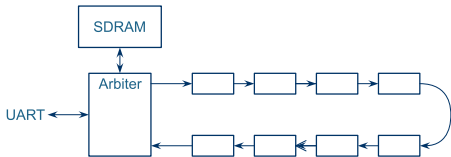
- ▶ When more than one module requests access, the one with higher priority always receives the grant
- ▶ Can result in starvation of low-priority modules
- ▶ Fast (same-cycle response by means of a combinational circuit)
- ▶ Scales well



Arbitration

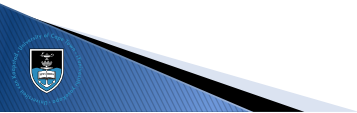
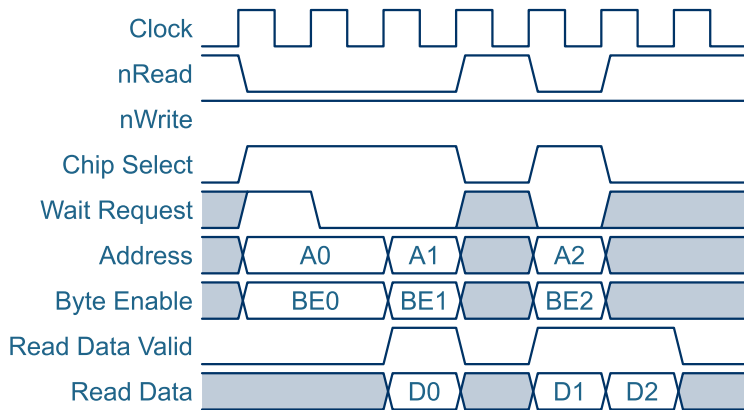


Arbitration



- ▶ The arbiter makes it look as if the resource has multiple independent interfaces. There are no control lines other than the interface itself.
- ▶ Often implemented by means of an embedded mutual exclusion unit (round-robin or priority based)
- ▶ The I²C standard implements arbitration by collision-detection and random back-off: the modules monitor their own output, and when there is a mismatch, the module waits and tries again later.

Avalon Bus Arbitration



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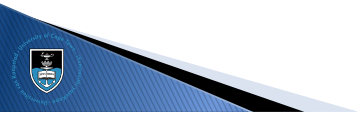
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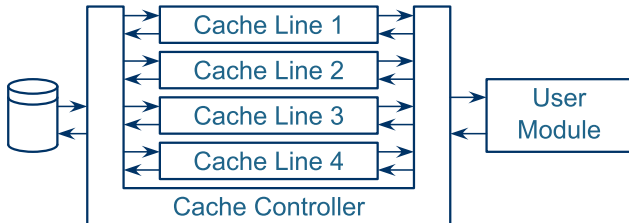
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General Cache



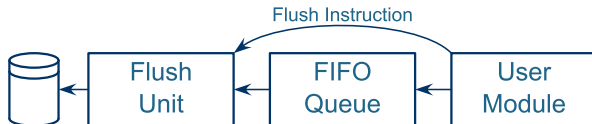
- ▶ Typical cache line sizes are 32, 64, 128 or 256 bytes
- ▶ Controller does address translation between external and cache line addresses
- ▶ Read on-demand, flush the least recently used line and load the desired content
- ▶ Controller blocks the user module when a cache miss occurs

Pre-fetch Read Cache



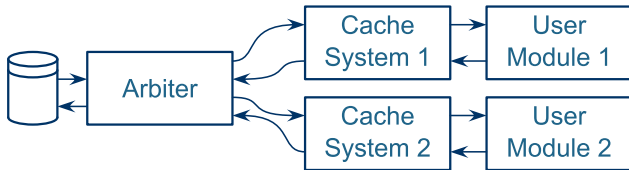
- ▶ Continuous reading: the user module is only blocked when the queue is empty
- ▶ The queue is typically multiple pages in size
- ▶ New data is fetched as soon as there is one page open on the queue

Write Cache



- ▶ Writing is continuous, and the user module is only blocked when the queue is full
- ▶ Alternatively, the user module is never blocked (data dropped on overflow)
- ▶ The flush unit automatically flushes the queue when there is one page of data
- ▶ The user module can also manually flush the queue if required

Cache Coherence



- ▶ Two independent cache system can refer to the same place in memory
- ▶ When the one user module reads what the other is writing, the two cache system must remain up to date with each other
- ▶ When the two user modules write to different places in the same page, a flush must not overwrite the other's data
- ▶ Many schemes exist to mitigate these and other issues, which are outside the scope of this course

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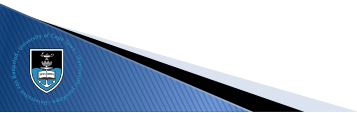
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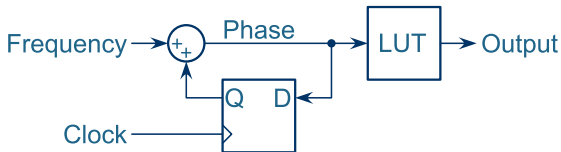
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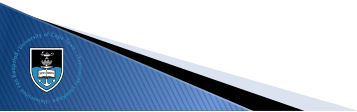
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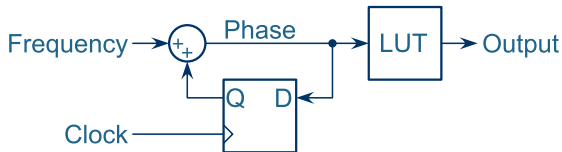
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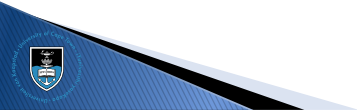
- ▶ Integrate frequency to obtain phase
- ▶ Use on-chip memory blocks as a look-up table to convert phase to the intended waveform
- ▶ Use dual-port ROM to obtain sine and cosine from the same LUT
- ▶ Higher resolution can be obtained with a CORDIC-based sine-cosine module



Numerically-controlled Oscillator



$$f_{out} = f_s \cdot \frac{f_{in}}{2^N}, \quad N = 24, \quad f_s = 100 \text{ MHz}$$



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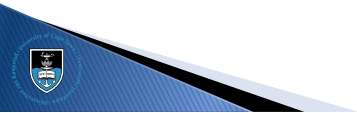
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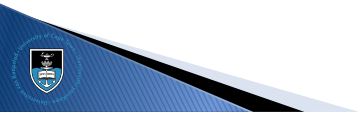
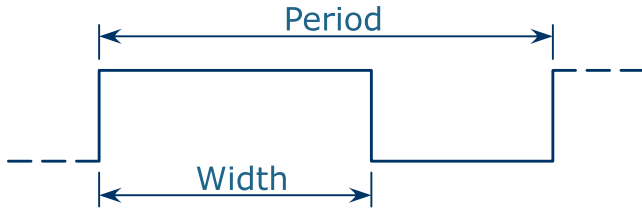
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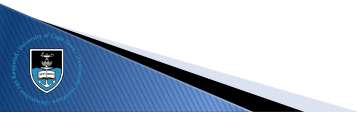
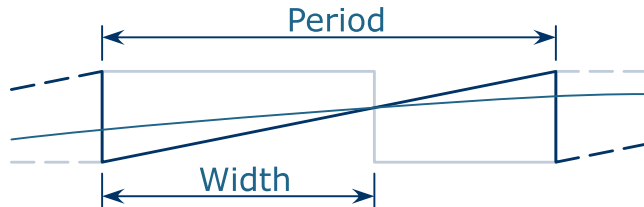
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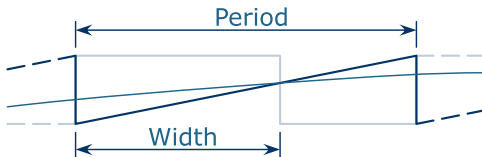
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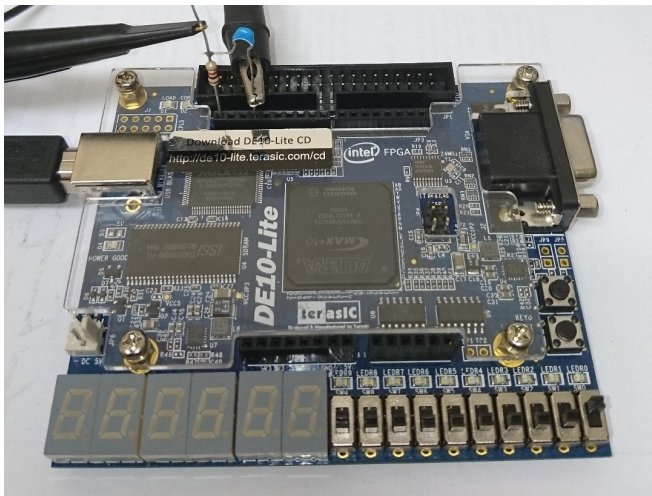


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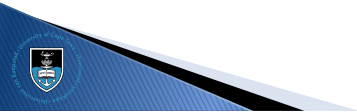
- ▶ Typically, the signal is in 2's complement
- ▶ The PWM module requires an unsigned input from 0 to $(2^N - 1)$ – also known as offset-binary
- ▶ To convert from one to the other, invert the most-significant bit

Pulse-width Modulation



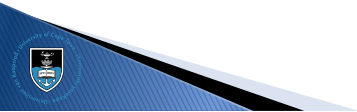
High Resolution PWM

- ▶ Assume that your signal is audio (20 kHz bandwidth)
- ▶ You want the PWM frequency at least 10 times the signal bandwidth $\Rightarrow$ 200 kHz
- ▶ Say you want 16-bit output resolution...
- ▶ You have to run the saw-tooth counter at $200 \text{ kHz} \times 2^{16} = 13.1072 \text{ GHz}$
- ▶ That is a FAST clock!
- ▶ The solution...



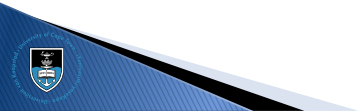
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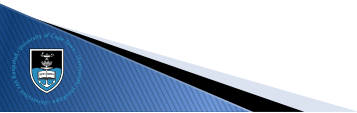
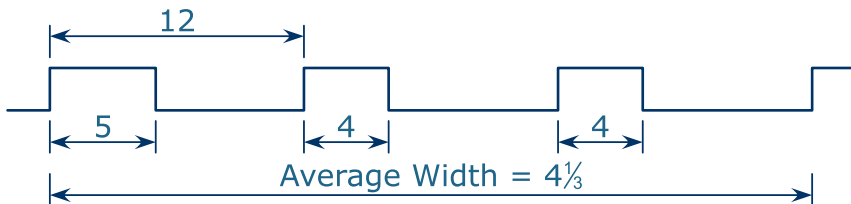


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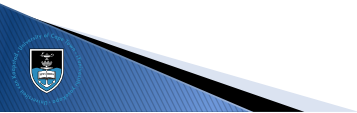
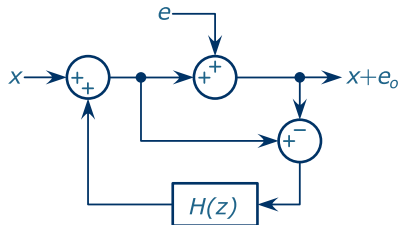
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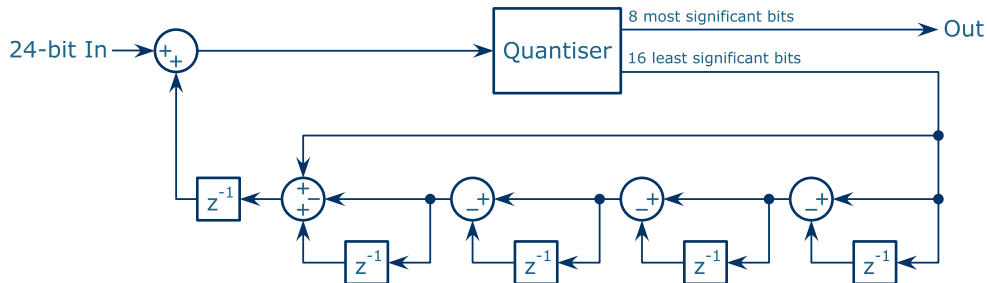
Noise Shaping



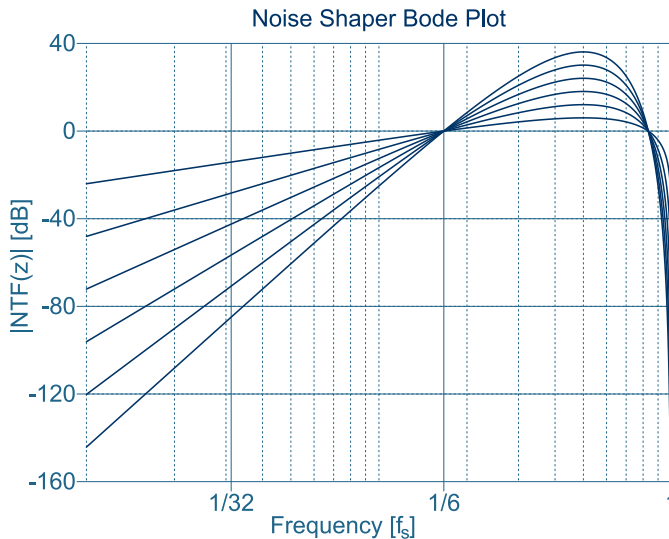
Noise Shaping



Noise Shaping



Noise Shaping



Select References



Stephen Brown and Zvonko Vranesic
Fundamentals of Digital Logic with Verilog Design, 2nd Edition
ISBN 978-0-07-721164-6



Merrill L Skolnik
Introduction to RADAR Systems
ISBN 978-0-07-288138-7



Mark A. Richards and James A. Scheer
Principles of Modern Radar: Basic Principles
ISBN 978-1-89-112152-4



Deepak Kumar Tala
World of ASIC
<http://www.asic-world.com/>



Jean P. Nicolle
FPGA 4 Fun
<http://www.fpga4fun.com/>



FPGA Development for Radar, Radio-Astronomy and Communications

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Dept. Electrical Engineering, University of Cape Town
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